

Database Architecture — Best Invest Properties

Date: 25 September 2026

Storage: Bubble database

Participating systems: Bubble, Make, OpenAI API

1. Purpose and principles

The model must support the three real MVP loops:

- an investor finds, compares, saves and enquires about a specific available unit;
- a developer gets verified, submits a project and maintains price/availability;
- an admin checks documents, publishes content and controls introductions, the score and the AI narrative.

There is one database — Bubble. It holds **30 data types** and **23 Option Sets**.

Principles:

1. **Bubble is the source of truth.** The Make Data Store is not a business database.
2. **Unit is the public listing object.** Project describes the building/complex, Unit Type the repeated configuration, Unit the specific offer.
3. **A single property does not get its own table.** It is a project with one type and one unit.
4. **Finance is deterministic.** AI never writes price, rent, yield, score or tax.
5. **History only where it is needed:** the score model and scores, user consents, and the sources of financial figures. Other changes overwrite the value, and the Audit Event records who changed it and when.
6. **Privacy by default.** New types are created private; only whitelisted fields of published listings are public.
7. **Minimal denormalisation for Bubble.** Fields needed by privacy rules and frequent searches are duplicated on the protected record.
8. **Idempotent integrations.** Every asynchronous process has an Integration Job and a unique idempotency key.

Bubble privacy rules run on the server and must be the primary mechanism that keeps data out of the browser. Visually hiding a group or button is not access control.

2. Conceptual schema

```
erDiagram
    USER ||--o| INVESTOR_PROFILE : has
    DEVELOPER_COMPANY ||--o{ USER : employs
    DEVELOPER_COMPANY ||--o{ DEVELOPER_APPLICATION : submits
    DEVELOPER_APPLICATION ||--o{ VERIFICATION_DOCUMENT : includes
    USER ||--o{ CONSENT_RECORD : grants

    COUNTRY_CONFIG ||--o{ COST_RULE : has
    COUNTRY_CONFIG ||--o{ DOCUMENT_REQUIREMENT : has
    COUNTRY_CONFIG ||--o{ PROJECT : contains
    DEVELOPER_COMPANY ||--o{ PROJECT : owns
    PROJECT ||--o{ UNIT_TYPE : defines
    UNIT_TYPE ||--o{ UNIT : instantiates
    PROJECT ||--o{ MEDIA_ASSET : has

    UNIT ||--o{ FINANCIAL_INPUT : has
    RENTAL_COMPARABLE_SET ||--o{ FINANCIAL_INPUT : sources
    UNIT ||--o{ ANALYSIS_FACT : has

    SCORE_MODEL_VERSION ||--o{ LISTING_SCORE : produces
    UNIT ||--o{ LISTING_SCORE : receives
    LISTING_SCORE ||--o{ SCORE_COMPONENT : contains
    LISTING_SCORE ||--o{ SCORE_SUB_COMPONENT : contains
    LISTING_SCORE ||--o{ AI_ANALYSIS : explained_by

    USER ||--o{ SAVED_UNIT : saves
    USER ||--o{ SAVED_SEARCH : owns
    USER ||--o{ CALCULATOR_SCENARIO : owns
    USER ||--o{ ENQUIRY : creates
    UNIT ||--o{ ENQUIRY : concerns
    ENQUIRY ||--o{ ENQUIRY_STATUS_EVENT : has

    UNIT ||--o{ CHANGE_REQUEST : receives
    INTEGRATION_JOB ||--o{ AUDIT_EVENT : emits
```

Market Benchmark is tied to a country and district rather than to a specific unit, so it is not shown separately on the diagram.

3. Naming rules in Bubble

- Data types: singular English, Pascal Case: `Developer Company`, `Score Model Version`.
- Fields: snake_case English: `publication_status`, `price_eur`.
- Option sets: Pascal Case: `Project Status`; options are stable lower-case machine keys.
- Every business type has `public_id` (text for URLs/references), and `is_archived` where the record can be archived.

- The Bubble `unique_id` is used internally only; `public_id` is what goes outside.
- Money in the MVP: numeric fields with the `_eur` suffix; the currency is always EUR.
- Percentages: store the decimal fraction (`0.0725`), display as `7.25%`.
- Time: all timestamps in UTC; timezone is used only for display.
- Bubble adds `Created Date`, `Modified Date` and `Creator` automatically — they are not repeated in the tables below.

4. Option Sets

Option Sets are for fixed, rarely changing, non-secret values. Bubble states explicitly that option sets are part of the app code, are not protected by privacy rules and are unsuitable for sensitive data. Changing them requires a deploy.

Option Set	MVP values
User Role	<code>investor</code> , <code>developer</code> , <code>admin</code>
Account Status	<code>pending_email</code> , <code>active</code> , <code>suspended</code> , <code>closed</code>
Application Status	<code>draft</code> , <code>submitted</code> , <code>under_review</code> , <code>more_info_required</code> , <code>approved</code> , <code>rejected</code> , <code>withdrawn</code>
Project Status	<code>draft</code> , <code>submitted</code> , <code>under_review</code> , <code>changes_requested</code> , <code>approved</code> , <code>published</code> , <code>paused</code> , <code>sold_out</code> , <code>archived</code> , <code>rejected</code>
Publication Status	<code>not_published</code> , <code>published</code> , <code>hidden</code> , <code>archived</code>
Unit Availability	<code>available</code> , <code>reserved</code> , <code>sold</code> , <code>withdrawn</code>
Financial Input Source	<code>developer_claim</code> , <code>portal_comparable</code> , <code>analyst_verified</code> , <code>platform_default</code> , <code>calculated</code>
Analysis Fact Tag	<code>source</code> , <code>developer</code> , <code>estimate</code> , <code>gap</code>
Change Request Status	<code>submitted</code> , <code>queried</code> , <code>approved</code> , <code>rejected</code> , <code>applied</code> , <code>cancelled</code>
Enquiry Status	<code>submitted</code> , <code>screening</code> , <code>on_hold</code> , <code>declined</code> , <code>approved_for_intro</code> , <code>introduced</code> , <code>developer_responded</code> , <code>qualified</code> , <code>closed_won</code> , <code>closed_lost</code> , <code>cancelled</code>
Score Verdict	<code>strong</code> , <code>good</code> , <code>watch</code> , <code>high_risk</code> , <code>not_scored</code>
Review Status	<code>not_required</code> , <code>pending</code> , <code>approved</code> , <code>changes_requested</code> , <code>rejected</code>
Analysis Status	<code>queued</code> , <code>generating</code> , <code>generated</code> , <code>failed</code> , <code>stale</code> , <code>published</code>
Job Status	<code>queued</code> , <code>processing</code> , <code>succeeded</code> , <code>retry_wait</code> , <code>failed</code> , <code>dead_letter</code> , <code>cancelled</code>
Document Kind	<code>company_registration</code> , <code>licence</code> , <code>ownership</code> , <code>planning</code> , <code>title</code> , <code>brochure</code> , <code>floor_plan</code> , <code>legal</code> , <code>other</code>
Financing Mode	<code>cash</code> , <code>mortgage</code>
Strategy	<code>long_term_rental</code> , <code>short_term_rental</code> , <code>capital_growth</code> , <code>mixed</code>

Option Set	MVP values
Score Criterion	5 criteria — see below
Score Sub-criterion	13 sub-criteria — see below
Amenity	communal_pool, gym, gated_area, underground_parking, tennis_golf, concierge, lift, landscaped_gardens
Data Provider	approved portals / market-data providers; attributes: name, country, access_method (api / feed / manual), terms_reference, fetch_frequency_days, is_active
Media Kind	photo, floor_plan, brochure, document
Consent Type	terms, privacy, marketing, contact_sharing

What is not an Option Set. Country Config, Cost Rule, Document Requirement and Market Benchmark are data types, because the admin edits them from the admin area without a deploy (screen A11). Legal texts (Terms, Privacy, Cookie, disclaimers) are static pages; the version the user agreed to is stored on the Consent Record.

Unit Availability. A developer may set only available, reserved or sold, and every such change goes through a Change Request with approval. Only an admin sets withdrawn. This must be enforced in the backend workflow whitelist, not in the UI.

Score Criterion — attributes

key	label	max_points	default_weight	Note
income	Rental Income & Net Yield	30	0.30	rating derived automatically from net yield bands
demand	Rental Demand & Tenant Quality	20	0.20	from sub-criteria
value	Purchase Value & Market Position	20	0.20	from the deviation of price per m² from the Market Benchmark
growth	Growth & Resale Potential	15	0.15	from sub-criteria
risk	Risk & Investor Protection	15	0.15	from sub-criteria; more points = lower risk, scale label mandatory in the UI

Each criterion gets a rating of 0–10, then $\text{weighted_points} = \text{rating} / 10 \times \text{weight}$. The full scales are in FR-03b of the project specification.

Score Sub-criterion — attributes

criterion	key	label	max_points
demand	market_activity	Market activity and number of comparable listings	3
demand	year_round	Year-round demand	3
demand	tenant_mix	Diversity of potential tenants	2
demand	seasonality	Seasonality and vacancy risk	2

critterion	key	label	max_points
growth	price_trend	Price trend	4
growth	liquidity	Liquidity and number of transactions	3
growth	infrastructure	Infrastructure and economic factors	2
growth	data_quality	Recency and completeness of data	1
risk	legal_title	Legal status / title	3
risk	permits	Building permits and documents	2
risk	payment_protection	Payment / escrow protection	2
risk	developer_check	Developer check	2
risk	stage_risk	Construction stage risk	1

The `max_points` of each criterion's sub-criteria add up to **10**.

Role-specific Enquiry status labels

The eleven `Enquiry Status` values are **canonical**. Role labels are three display attributes of the same Option Set (`label_admin` , `label_investor` , `label_developer`), not additional statuses.

Canonical status	Admin	Investor	Developer
submitted	New	Request received	not shown
screening	Qualifying	Awaiting analysis	not shown
qualified	Hot lead / Awaiting you	Awaiting admin approval	Awaiting admin approval
on_hold	On hold	Under review	On hold
declined	Declined	Not available	Filtered out by Best Invest
approved_for_intro	Connecting	Introduction being prepared	Introduction being prepared
introduced	Connected	Developer contacted	Introduced
developer_responded	In progress	Developer responded	Contacted
closed_won	Converted	Completed	Converted
closed_lost	Closed	Closed	Closed
cancelled	Cancelled	Cancelled	Cancelled

An empty label means the record is not shown to that role at all — this is enforced by a privacy rule, not just hidden in the UI.

AWAITING YOU is not a status. It is the admin label meaning a lead in `qualified` is waiting for their decision. Likewise, "Hot lead" is a label for the same status, not a separate stage.

5. Data types

30 types in six groups.

Group	Types
5.1 People and access	User, Investor Profile, Developer Company, Developer Application, Verification Document, Consent Record
5.2 Market settings	Country Config, Cost Rule, Document Requirement, Market Benchmark
5.3 Catalogue	Project, Unit Type, Unit, Media Asset
5.4 Finance, rent, score	Financial Input, Rental Comparable Set, Analysis Fact, Score Model Version, Listing Score, Score Component, Score Sub-component, AI Analysis
5.5 Investor and enquiries	Saved Unit, Saved Search, Calculator Scenario, Enquiry, Enquiry Status Event, Change Request
5.6 System	Integration Job, Audit Event

5.1 People and access

User (Bubble built-in)

Field	Type	Required	Note
public_id	text	yes	usr_..., not the email
full_name	text	yes	do not use as an identifier
phone_e164	text	no	private
country_of_residence	text	no	private
roles	list of User Role	yes	at least one role after activation
account_status	Account Status	yes	route guard
email_verified	yes/no	yes	false by default
developer_company	Developer Company	no	for developers; one user per company in the MVP
admin_permission_keys	list of text	no	admin only; permissions are checked by key, not just by role
last_login_at	date	no	audit

Do not store the password in a custom field. Use Bubble authentication.

Investor Profile

Field	Type	Required	Note
user	User	yes	one active profile per User
budget_min_eur	number	no	
budget_max_eur	number	no	
target_gross_yield	number	no	decimal fraction
target_net_yield	number	no	decimal fraction
time_horizon_months	number	no	

Field	Type	Required	Note
countries	list of Country Config	no	preferred countries
strategies	list of Strategy	no	
bedrooms_min	number	no	
marketing_email_opt_in	yes/no	yes	mirrors the latest marketing consent for fast filtering
match_email_frequency	text	no	off / daily / weekly

Profile uniqueness is checked by a backend workflow before creation.

Developer Company

Field	Type	Required	Note
public_id	text	yes	
legal_name	text	yes	
registration_number	text	yes	
registration_country	Country Config	yes	
licence_number	text	yes	
website_url	text	no	
years_active	number	no	
projects_completed	number	no	
projects_selling	number	no	
typical_unit_price_eur	number	no	
verification_status	Application Status	yes	mirrored from the latest application for privacy rules
verified_at	date	no	
is_suspended	yes/no	yes	

The Company owns the Project. Never link a Project directly to a single developer User.

Developer Application

Field	Type	Required	Note
public_id	text	yes	APP-0142
company	Developer Company	yes	
submitted_by_user	User	yes	
status	Application Status	yes	
markets	list of Country Config	yes	
submitted_at	date	no	
decided_at	date	no	

Field	Type	Required	Note
decision_reason_private	text	no	admin only
decision_message_public	text	no	visible to the developer

Verification Document

Field	Type	Required	Note
application	Developer Application	yes	
company	Developer Company	yes	duplicated for privacy rules
document_kind	Document Kind	yes	
file	file	yes	always private
expires_at	date	no	
verification_status	Review Status	yes	
review_note_public	text	no	visible to the developer
reviewed_at	date	no	
is_current	yes/no	yes	a re-upload sets the previous one to no

The file is always private and attached to the Verification Document. Removing the URL does not delete the file; the delete workflow must first run the Bubble action "Delete an uploaded file".

Consent Record

Field	Type	Required	Note
user	User	yes	
consent_type	Consent Type	yes	
granted	yes/no	yes	
document_version	text	yes	version of the text the user agreed to, e.g. terms-2026-09
source_page	text	no	where consent was given
withdrawn_at	date	no	

Consent is never overwritten: every change is a new record. This is a GDPR requirement — you must be able to prove what exactly the user agreed to.

5.2 Market settings

Country Config

Field	Type	Required	Note
iso2	text	yes	CY , ES
name	text	yes	
is_live	yes/no	yes	
coming_soon	yes/no	yes	for Greece/Portugal if needed

Field	Type	Required	Note
default_vacancy_rate	number	yes	decimal fraction
default_management_fee_rate	number	yes	
default_maintenance_rate	number	yes	
default_insurance_annual_eur	number	yes	
legal_disclaimer	text	no	
min_comparable_listings	number	yes	default 5 — below this the sample cannot be approved
sufficient_comparable_listings	number	yes	default 10 — below this the sample is approved with a "thin sample" flag
max_comparable_age_days	number	yes	default 90 — after this the sample is stale

Sample thresholds are stored per country because market liquidity differs: Nicosia has fewer listings than Málaga. A median of four listings is not a market figure — one outlier moves it too much, so 5 is the absolute minimum. From 10 listings the median is stable. 90 days is a quarter: no need to re-collect every week, and the estimate keeps up with the market. The admin changes all values without code changes.

Cost Rule

Field	Type	Required	Note
country	Country Config	yes	
rule_key	text	yes	transfer_tax, stamp_duty, legal_fees, ...
label	text	yes	
applies_to_new_build	yes/no	no	empty = applies to all
calculation_method	text	yes	flat / percent_price / banded
rate	number	no	for percent_price
fixed_amount_eur	number	no	for flat
bands_json	text	no	for banded
source_url	text	yes	official source
source_checked_at	date	yes	
approved_by	text	no	who checked the rate (consultant)

Do not store all tax rules as one text. Edited from the admin area (A11); the value is overwritten and the Audit Event records the change history.

Document Requirement

Field	Type	Required	Note
country	Country Config	yes	
subject_type	text	yes	developer / project
document_kind	Document Kind	yes	
required_for_account_creation	yes/no	yes	e.g. company registration, licence
required_for_submission	yes/no	yes	
required_for_publication	yes/no	yes	e.g. building permit, escrow
expiry_months	number	no	empty = no expiry
instructions	text	no	hint for the developer
is_active	yes/no	yes	

The three `required_for_*` fields express the three levels on the application screen: `REQUIRED` / `TO CONFIRM` / `OPTIONAL`. The matrix is **fully configurable** by the admin per country and is not hard-coded.

Market Benchmark

Needed for the `Purchase Value & Market Position` criterion, which compares the price per m² with the district median.

Field	Type	Required	Note
country	Country Config	yes	
city	text	yes	
area_name	text	no	empty = whole city
property_type	text	no	
median_price_per_m2_eur	number	yes	median for new builds
sample_size	number	no	number of properties in the sample
source_reference	text	yes	public listings / transaction register
checked_at	date	yes	date checked

5.3 Property catalogue

Project

Field	Type	Required	Note
public_id	text	yes	PRJ-0311
company	Developer Company	yes	never returned to an anonymous request
country	Country Config	yes	
city	text	yes	
area_name	text	no	
address_private	text	no	not published

Field	Type	Required	Note
map_lat_rounded	number	no	rounded for the public map
map_lng_rounded	number	no	
name	text	yes	
project_type	text	yes	single / complex
description_public	text	no	
completion_date	date	yes	
declared_unit_count	number	yes	
amenities	list of Amenity	no	
cover_media	Media Asset	no	required for publication
status	Project Status	yes	
publication_status	Publication Status	yes	
published_at	date	no	

Country/city/area are enough for search cards; `address_private` is never exposed publicly.

Developer anonymity. The developer's name and contacts are not shown in the catalogue or on the property page; the public label is "Introduced by Best Invest". The privacy rule on Project does not return `company` to an anonymous request at all — do not rely on the UI not showing it.

Unit Type

Field	Type	Required	Note
public_id	text	yes	
project	Project	yes	
name	text	yes	e.g. "Type B — 2 bed"
bedrooms	number	yes	
bathrooms	number	yes	
indoor_area_m2	number	yes	
outdoor_area_m2	number	no	
floor_plan	Media Asset	no	

Unit

The central listing record. It also holds the current financial metrics — there is no separate type for them.

Field	Type	Required	Note
public_id	text	yes	URL and external references
project	Project	yes	
unit_type	Unit Type	yes	
unit_number	text	yes	may be hidden until introduction

Field	Type	Required	Note
floor	number	no	
price_eur	number	yes	current approved price
availability	Unit Availability	yes	
has_pending_change	yes/no	yes	a Change Request is in progress — the portal row shows "pending approval"
monthly_rent_eur	number	no	approved rent used for scoring
purchase_costs_eur	number	no	calculated from Cost Rules
annual_net_income_eur	number	no	calculated
gross_yield	number	no	calculated
net_yield	number	no	calculated
financials_calculated_at	date	no	date of the last calculation
investment_score	number	no	current score
score_verdict	Score Verdict	no	badge
current_score	Listing Score	no	components and model version
country_cached	Country Config	yes	for search and privacy without a deep chain
city_cached	text	yes	search
bedrooms_cached	number	yes	search
area_m2_cached	number	yes	search
property_type_cached	text	yes	search
completion_date_cached	date	yes	completion filter
strategy_keys	list of Strategy	no	search
publication_status	Publication Status	yes	only <code>published</code> is public

Financial and cached fields are updated by the atomic backend workflow `recalculate_unit`. Do not build the public search over many nested relations.

Base MVP formulas:

```

annual_gross_rent = monthly_rent × 12
annual_effective_rent = annual_gross_rent × (1 - vacancy_rate)
gross_yield = annual_gross_rent / price
annual_net_income = annual_effective_rent
    - management - maintenance - insurance - property_tax - other_costs
net_yield = annual_net_income / (price + purchase_costs)

```

Inputs are approved Financial Inputs (rent comes from comparable listings via Make), plus the Country Config settings and Cost Rules for the country. The values used for a particular score are recorded on the Listing Score.

Media Asset

Field	Type	Required	Note
project	Project	yes	
unit_type	Unit Type	no	for floor plans
kind	Media Kind	yes	
image	image	no	for photo
file	file	no	for PDFs/documents
is_private	yes/no	yes	legal documents — yes
is_cover	yes/no	yes	
sort_order	number	yes	
caption	text	no	

Public photos/brochures and private documents are different records with different privacy rules.

5.4 Finance, rent, score

Financial Input

The Project Review screen shows a panel where every figure has its own source, and three rent values side by side (claimed by the developer, Best Invest's comparable estimate, and the one actually used for scoring). So each input value is a separate record.

Field	Type	Required	Note
unit	Unit	yes	
input_key	text	yes	monthly_rent, recurring_costs, vacancy_allowance, ...
label	text	yes	label in the review
value_number	number	yes	
source	Financial Input Source	yes	
source_reference	text	no	"Larnaca district, 14 lettings"
comparable_set	Rental Comparable Set	no	required for portal_comparable
is_used_for_scoring	yes/no	yes	only one active value per input_key
review_status	Review Status	yes	pending until approved by the admin
approved_at	date	no	
superseded_by	Financial Input	no	a new value does not overwrite the old one

Rules:

- only values with `review_status = approved` reach the public listing;
- the Unit's financial metrics are calculated **only** from approved Financial Inputs;
- a value with `source = developer_claim` can never have `is_used_for_scoring = yes` without separate analyst approval (BR-02);
- `source = portal_comparable` requires a `comparable_set` with `review_status = approved`;
- for rent there are always at least two records: the developer's claim (`developer_claim`) and the value derived from comparable listings (`portal_comparable`) — these are shown side by side on Project Review;
- `developer_claim` never turns into `portal_comparable` automatically;
- a record is not edited: a new value creates a new record and sets `superseded_by` on the old one. This shows where every published figure came from.

Rental Comparable Set

The result of collecting comparable rental listings via Make. One record = one sample for one parameter set on a given date.

Field	Type	Required	Note
public_id	text	yes	
provider	Data Provider	yes	Option Set
retrieved_at	date	yes	date the data was retrieved
country	Country Config	yes	normalisation parameter
city	text	yes	normalisation parameter
area_name	text	no	district
property_type	text	yes	normalisation parameter
bedrooms	number	yes	normalisation parameter
area_m2_min	number	no	sample floor-area bound
area_m2_max	number	no	sample floor-area bound
listings_count	number	yes	number of comparable listings
rent_min_eur	number	yes	lower end of the range
rent_median_eur	number	yes	median — basis of the base estimate
rent_max_eur	number	yes	upper end of the range
collection_method	text	yes	<code>api</code> / <code>feed</code> / <code>manual_import</code> / <code>manual_entry</code>
integration_job	Integration Job	no	empty for manual entry
review_status	Review Status	yes	<code>pending</code> until checked by the admin
is_current	yes/no	yes	current sample for this parameter set

Rules:

- a sample with `review_status ≠ approved` cannot be a source for a Financial Input and is not shown to the investor;

- `listings_count` below `Country Config.min_comparable_listings` (default **5**) **blocks approval**;
- `listings_count` between the minimum and `sufficient_comparable_listings` (default **10**) is approved with a **thin sample** flag: shown to the admin on review and added for the investor as an Analysis Fact tagged `estimate`;
- a sample older than `max_comparable_age_days` (default **90**) gets `is_current = no` and cannot be the source of a new Financial Input; published values do not disappear but are queued for refresh;
- a new sample does not overwrite the previous one: the old one gets `is_current = no`;
- a manual fallback (`manual_import` / `manual_entry`) is stored with the same type and fields, so the provenance of the figure stays visible.

The legal basis for using each portal's data (API terms, data licence) is checked before connecting it. The link to the terms is the `terms_reference` attribute in the Data Provider Option Set.

Analysis Fact

The "Sources, assumptions & gaps" block on the analysis and review screens.

Field	Type	Required	Note
unit	Unit	yes	
tag	Analysis Fact Tag	yes	
body	text	yes	fact text
source_url	text	no	
sort_order	number	yes	
is_active	yes/no	yes	

The `gap` tag marks a known absence of data and must be visible to the investor — it is a legally significant part of the disclosure.

Score Model Version

Field	Type	Required	Note
version_number	number	yes	v1, v2, ...
status	text	yes	<code>draft</code> / <code>active</code> / <code>retired</code>
weight_income	number	yes	decimal fraction, default 0.30
weight_demand	number	yes	default 0.20
weight_value	number	yes	default 0.20
weight_growth	number	yes	default 0.15
weight_risk	number	yes	default 0.15
change_reason	text	yes	shown in the versions table
properties_rescored	number	no	filled in after recalculation
activated_at	date	no	

Weight rules:

- the five weights of the active model add up to `1.0` ; until they do, Save is blocked (checked by a backend workflow);
- the editor works in percentages, 0-100 per criterion; there is **no** hard cap on a single criterion;
- the UI shows a soft warning if any criterion exceeds **50%** — one criterion would then effectively determine the whole score. The warning does not block saving; the threshold is a UI constant;
- the weights are the same for Cyprus and Spain;
- `Revert` creates a **new** version with the previous parameters rather than deleting history.

Listing Score

Field	Type	Required	Note
unit	Unit	yes	
model_version	Score Model Version	yes	
total_score	number	yes	0-100
verdict	Score Verdict	yes	
price_eur_used	number	yes	price at calculation time
monthly_rent_eur_used	number	yes	rent at calculation time
net_yield_used	number	yes	
gross_yield_used	number	yes	
calculated_at	date	yes	
is_current	yes/no	yes	a new calculation sets the previous one to <code>no</code>

The `*_used` fields record the inputs the score was calculated from, so the score stays reproducible even after the price or rent on the Unit changes.

Score Component

Field	Type	Required	Note
listing_score	Listing Score	yes	
criterion	Score Criterion	yes	Option Set
raw_value	number	no	e.g. net yield or deviation from the median; empty for assessed criteria
rating	number	yes	0-10
weighted_points	number	yes	<code>rating / 10 × weight × 100</code>
explanation	text	no	deterministic explanation for the analysis screen

Each Listing Score has exactly five components; they add up to the total score within `0.01`.

Score Sub-component

The actual assessment of a sub-criterion for a specific Listing Score.

Field	Type	Required	Note
listing_score	Listing Score	yes	
sub_criterion	Score Sub-criterion	yes	Option Set
points_awarded	number	yes	no more than the sub-criterion's <code>max_points</code>
note	text	no	analyst comment

The sum of `points_awarded` within a criterion gives its `rating`.

AI Analysis

Field	Type	Required	Note
unit	Unit	yes	
listing_score	Listing Score	yes	the score the text is based on
status	Analysis Status	yes	
review_status	Review Status	yes	
prompt_version	text	yes	
model_id	text	yes	
summary	text	no	
strengths_json	text	no	
risks_json	text	no	
missing_data_json	text	no	
input_tokens	number	no	
output_tokens	number	no	
reviewed_at	date	no	
published_at	date	no	
failure_code	text	no	

Only `review_status = approved` is published, and only while the linked Listing Score is still `is_current = yes`.

5.5 Investor and enquiries

Saved Unit

Field	Type	Required	Note
user	User	yes	
unit	Unit	yes	
price_at_save_eur	number	yes	for the "price changed" badge
availability_at_save	Unit Availability	yes	for the "status changed" badge

Field	Type	Required	Note
is_active	yes/no	yes	check for an active duplicate before creating

Saved Search

Field	Type	Required	Note
public_id	text	yes	
user	User	yes	
name	text	yes	
countries	list of Country Config	no	
property_types	list of text	no	
bedrooms	list of text	no	studio , 1 , 2 , 3+
strategies	list of Strategy	no	
completion	list of text	no	ready , lt_12m , 12_24m
price_max_eur	number	no	
gross_yield_min	number	no	
net_yield_min	number	no	
alert_frequency	text	yes	off / daily / weekly
is_active	yes/no	yes	

Calculator Scenario

Field	Type	Required	Note
public_id	text	yes	
user	User	yes	anonymous calculations are not stored
unit	Unit	yes	
name	text	yes	
strategy	Strategy	yes	long / short term
rent_scenario	text	yes	base / average / best
financing_mode	Financing Mode	yes	
purchase_price_eur	number	yes	
deposit_eur	number	no	for mortgage
interest_rate	number	no	for mortgage
term_years	number	no	for mortgage
monthly_rent_eur	number	yes	
occupancy_rate	number	yes	
management_fee_rate	number	yes	
annual_net_income_eur	number	yes	result at save time
cash_on_cash_return	number	no	result at save time

Enquiry

The contact release fields are stored right here — there is no separate type for them.

Field	Type	Required	Note
public_id	text	yes	ENQ-YYYY-... / for the developer LEAD-0412
investor_user	User	yes	private
unit	Unit	yes	the specific subject of the enquiry
project	Project	yes	denormalised
developer_company	Developer Company	yes	denormalised for privacy rules
status	Enquiry Status	yes	canonical status
budget_eur	number	no	value at submission time
consent_to_share_contact	yes/no	yes	separate consent
fit_summary_private	text	no	admin only
follow_up_date	date	no	required for on_hold
submitted_at	date	yes	SLA start
contact_released_at	date	no	filled in on Approve & connect
released_investor_name	text	no	copy the developer sees after release
released_investor_email	text	no	
released_investor_phone	text	no	
released_developer_contact	text	no	developer contact the investor sees
developer_outcome	text	no	lead outcome (D11): viewing_booked / reserved / not_interested
closed_at	date	no	

The developer **never** gets access to the investor's User record. Before release the `released_*` fields are empty; the privacy rule returns them to the developer only once `contact_released_at` is filled in.

Enquiry Status Event

Field	Type	Required	Note
enquiry	Enquiry	yes	
from_status	Enquiry Status	no	
to_status	Enquiry Status	yes	
actor_user	User	no	empty for automatic transitions
reason_private	text	no	required for on_hold / declined ; never shown to the investor

Field	Type	Required	Note
note_investor	text	no	what the investor sees in the history (P12)

A status never changes without a Status Event. These records are what the investor sees as the status history.

Change Request

One request = one change to one unit, or a content review request.

Field	Type	Required	Note
public_id	text	yes	
project	Project	yes	
unit	Unit	no	empty for content review
company	Developer Company	yes	for privacy rules
request_type	text	yes	price / availability / content_review
old_price_eur	number	no	
new_price_eur	number	no	
old_availability	Unit Availability	no	
new_availability	Unit Availability	no	only available / reserved / sold
content_message	text	no	for content_review: what the developer wants to change
status	Change Request Status	yes	
decision_reason	text	no	required for queried / rejected ; visible to the developer
submitted_at	date	yes	
decided_at	date	no	

After publication the developer changes only price and availability; both go through **prior** approval. Locked fields (project name, location, completion date, unit mix, specification, media) change through `content_review`. The Change Request is kept after it is applied — this is the change history on screen D08.

5.6 System

Integration Job

Every asynchronous action through Make, including every email.

Field	Type	Required	Note
public_id	text	yes	correlation id
job_type	text	yes	ai_analysis_generate, email, introduction, rental_comparables, score_rescore_batch, ...

Field	Type	Required	Note
entity_type	text	yes	
entity_public_id	text	yes	
recipient_user	User	no	for emails
template_key	text	no	for emails
status	Job Status	yes	
idempotency_key	text	yes	unique
attempt_count	number	yes	
processed_count	number	no	batch recalculation progress
last_error_redacted	text	no	no PII or secrets
provider_message_id	text	no	email id / OpenAI response id
completed_at	date	no	

Uniqueness is enforced by `idempotency_key`. Make reads the Job first; if it is `succeeded`, the repeat ends without side effects. This same type is the log of sent emails.

Audit Event

Field	Type	Required	Note
actor_user	User	no	empty for system actions
action_key	text	yes	<code>project.published</code> , <code>cost_rule.updated</code> , ...
entity_type	text	yes	
entity_public_id	text	yes	
before_json	text	no	no PII
after_json	text	no	no PII
reason	text	no	
source	text	yes	<code>bubble_ui</code> / <code>bubble_backend</code> / <code>make</code> / <code>openai</code>

Audit Event is append-only; user workflows may not change or delete records. It records who changed settings (rates, thresholds, documents) and when, instead of keeping separate versions of each record.

6. Status transitions

Project lifecycle

```
draft → submitted → under_review
under_review → changes_requested → submitted
under_review → approved → published
under_review → rejected
published → paused | sold_out | archived
```

The admin performs `approved → published` as a separate action. Publishing is not allowed without approved verification, at least one available unit, a cover photo, calculated financials, a current score, and an approved AI analysis or an explicit deterministic-only fallback.

A developer can **submit** a project with an incomplete set of units (modal: "12 units declared · 3 entered. You can submit and add the rest before publication"). So unit completeness is a condition of `approved → published`, not of `draft → submitted`. In addition, publishing requires every Financial Input used in the score to have `review_status = approved`.

Enquiry decision

UI action	Canonical transition	Emails
Approve & connect	<code>qualified → approved_for_intro → introduced</code>	to both parties
Hold	<code>qualified → on_hold</code>	none
Decline	<code>qualified → declined</code>	investor only, neutral text without a reason

`on_hold` and `declined` require `reason_private`. The text the investor sees on a decline comes from a template and does **not** contain the reason. `on_hold` additionally requires `follow_up_date` on the Enquiry.

Developer Application lifecycle

```
draft → submitted → under_review
under_review → more_info_required → submitted
under_review → approved | rejected
submitted/under_review → withdrawn
```

Enquiry lifecycle

```
submitted → screening → qualified → approved_for_intro → introduced
screening | qualified → on_hold | declined
on_hold → qualified | declined
introduced → developer_responded → closed_won | closed_lost
submitted | screening | qualified → cancelled
```

`qualified` means "the lead has been checked by the analyst and is waiting for the admin's decision", so it comes **before** `approved_for_intro`. The stage after the developer responds is `developer_responded`, then `closed_won` / `closed_lost`.

7. Privacy rules matrix

Data type	Anonymous	Investor owner	Developer company member	Admin
User	none	own whitelisted fields	own only	full
Investor Profile	none	own	none	full
Developer Company	none	none	own	full
Developer Application / Verification Document	none	none	own company	full
Consent Record	none	own	own	full
Project	published public subset, without company	published public subset	own company	full
Unit	published + available public subset	same	own company units	full
Media Asset public	published records	published records	own company	full
Media Asset private	none	none	own company	full
Financial Input	none	none	own project, approved only	full
Rental Comparable Set	none	none	none	full
Analysis Fact / Listing Score / Score Component	published subset	published subset	own project	full
AI Analysis	approved/published only	approved/published only	own project published view	full
Enquiry	none	own, investor-safe fields	own company; released_* fields only after release	full
Enquiry Status Event	none	own, without reason_private	own company, without reason_private	full
Change Request	none	none	own company	full
Integration Job / Audit Event	none	none	none	full

Implementation details:

- privacy conditions check direct fields on the record itself; do not build multi-level chains;
- duplicate `company`, `publication_status` on protected types where that simplifies the rule;
- the workflow `only when` checks role, company, account status and the allowed transition; privacy rules do not replace workflow authorisation;
- do not allow auto-binding for critical fields;
- do not expose the Data API for business types publicly; Make calls narrow authenticated Workflow API endpoints.

8. Bubble workflows that maintain integrity

Workflow	Trigger	Result
create_developer_application	submit form	User + Company + Application without duplicates, consent, audit
submit_project	developer action	validation, status event, review
publish_project	admin action	server validation, publication, audit
store_rental_comparables	callback from MK-10	Rental Comparable Set pending , previous one → is_current = no
approve_rental_comparables	admin action	sample approved , becomes available as a Financial Input source
approve_financial_input	admin action on A04	Financial Input approved , previous one → superseded_by
recalculate_unit	approved price/rent/settings change	updated Unit financial and cached fields
calculate_unit_score	financials or model change	new Listing Score + components
queue_ai_analysis	current score created	Integration Job ai_analysis_generate
rescore_all_published	admin saved a new Score Model Version	batch recalculation with progress, a new Listing Score per unit, old ones → is_current = no
submit_enquiry	investor action	consent check, Enquiry + status event
approve_introduction	admin action	release fields on the Enquiry + delivery Integration Job
submit_change_request	developer action	Change Request submitted , has_pending_change = yes
apply_change_request	admin approval	apply the change once, recalc, audit
close_account	confirmed request	suspend access, queue export/anonymisation review

Use database triggers only as a safety net for changes from the Bubble editor/API. Main second-order updates are better called from the primary backend workflow, so workload is not spent checking every change.

9. Search and performance

- The screen 02 search runs over Unit, with constraints: publication_status , availability , country_cached , price_eur , bedrooms_cached , gross_yield , net_yield , strategy_keys , completion_date_cached .
- Do not use :filtered for the server-filtered catalogue; all main conditions must be search constraints.
- Debounce slider changes by 300-500 ms.
- Sorting by net_yield , gross_yield , price_eur , investment_score .
- Pagination: **8 records per page**, numbered pager.

- The "Top 5" rail takes the first five records of the same sorted set — no separate query is needed.
- Do not store record lists on User if they grow without bound; Saved Unit and Enquiry are separate types.

10. Retention and deletion

Retention periods are set in the platform's legal policy. MVP technical policy:

- account closure blocks login immediately; the actual delete/anonymise is a controlled backend process;
- open enquiries, consent, audit and transactional records are not deleted automatically before the legal hold ends;
- investor PII in closed records is replaced with a pseudonymous reference after the approved period;
- an uploaded file is deleted by a separate action before the URL is cleared;
- an Integration Job contains no full PII — only a reference to the record;
- the OpenAI input contains property facts but no investor contact data.

11. Seed data before build

1. Country Config: Spain, Cyprus (with sample thresholds); Greece/Portugal as `coming_soon` if needed.
2. Cost Rules for both countries — with source and check date.
3. `Data Provider` Option Set: approved portals for Spain and Cyprus.
4. Market Benchmark: median price per m² for the launch districts.
5. Document Requirements per country.
6. The first Score Model Version (30/20/20/15/15).
7. Email templates and role-specific Enquiry Status labels.

12. Migration between Bubble Development and Live

- deploy structure/option sets through Bubble version control;
- export/import settings (Country Config, Cost Rule, Document Requirement, Market Benchmark) in a defined order;
- do not copy Live PII into Development without a written basis and masking;
- OpenAI/Make credentials are separate for Development and Live;
- smoke dataset: 2 countries, 2 companies, 4 projects, 8 unit types, 20 units, all availability/status cases.

13. Architecture acceptance criteria

- no anonymous request receives an email, phone, exact private address or private document URL;
- before introduction the developer sees the lead without PII; after approval sees only the `released_*` fields;

- a repeated Make webhook with the same idempotency key creates no duplicate and sends no repeat email;
- every published unit has calculated financials, a current Listing Score and a Score Model Version;
- every number on Analysis is traceable to a Financial Input or a Listing Score;
- score components equal the total score within 0.01, and there are exactly five;
- a change to an approved price creates a Change Request, an Audit Event, a Unit recalculation and a new Listing Score;
- an availability change likewise goes through a Change Request and is not applied before approval;
- an anonymous request does not receive `Project.company` or any developer field;
- no Financial Input with `source = developer_claim` takes part in scoring without separate approval;
- every published unit has at least one Analysis Fact or an explicit note that there are no gaps;
- deleting a private record does not leave an accessible attached file;
- every status change creates a Status Event/Audit Event;
- admin publish and contact release are impossible without server-side authorisation.